



Volumes of Revolution · Worksheet 1

Core Pure 2 Chapter 4: Volumes of Revolution. Answer all questions. Show your working clearly. Total: 47 marks. Time: 60 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are two sections, one for each topic in Chapter 4. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- All rotations are through 2π radians. Give exact answers in terms of π unless told otherwise.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Volumes of revolution about the y-axis	21	
B Volumes of revolution for parametric curves	21	
Roar-level challenge	5	
Total	47	

Section A Volumes of revolution about the y-axis

21 marks

1. The region bounded by the curve $y = x^2$, $x \geq 0$, the y-axis and the line $y = 4$ is rotated about the y-axis. Find the exact volume of the solid formed. [2]

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2. The region bounded by the curve $y = \frac{2}{x}$, $x > 0$, the y-axis and the lines $y = 1$ and $y = 4$ is rotated about the y-axis. Find the exact volume of the solid formed. [3]

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3. The region bounded by the curve $y = x^3$, $x \geq 0$, the y -axis and the lines $y = 1$ and $y = 8$ is rotated about the y -axis. Find the exact volume of the solid formed. [3]

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4. The region R is enclosed between the curve $y = x^2$ and the line $y = 2x$. The region R is rotated about the y -axis. Find the exact volume of the solid formed. [4]

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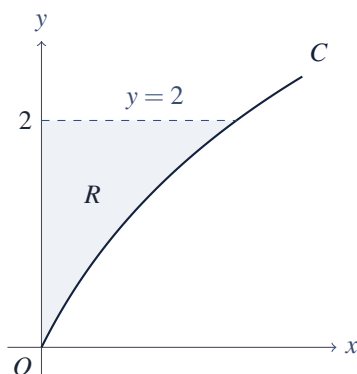
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5. The curve C has equation $y = 2\ln(x+1)$, $x \geq 0$. The region R is bounded by C , the y -axis and the line $y = 2$, as shown below.



The region R is rotated about the y -axis to form a solid S .

- (a) Show that the volume of S is given by $\pi \int_0^2 \left(e^{\frac{y}{2}} - 1\right)^2 dy$. [2]

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- (b) Hence find the exact volume of S . [4]

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- (c) The solid S is used to model a glass paperweight, where 1 unit on each axis represents 5 cm. [3]
The glass has density 2.5 g per cm^3 . Find, according to the model, the mass of the paperweight, giving your answer in kg to 3 significant figures.

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(Total for Question 5 is 9 marks)

**Section B Volumes of revolution for parametric curves**

21 marks

6. The curve C has parametric equations $x = 2t$, $y = t^2$, $0 \leq t \leq 2$. The region bounded by C , the x -axis and the line $x = 4$ is rotated about the x -axis. Show that the volume of the solid formed is $\frac{64\pi}{5}$. [2]

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7. The curve C has parametric equations $x = t^2$, $y = 2t$, $t \geq 0$. The region bounded by C , the x -axis and the lines $x = 1$ and $x = 9$ is rotated about the x -axis. Find the exact volume of the solid formed. [3]

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8. The curve C has parametric equations $x = t^2$, $y = t^3$, $0 \leq t \leq 1$. The region bounded by C , the y -axis and the line $y = 1$ is rotated about the y -axis. Find the exact volume of the solid formed. [3]

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9. The curve C has parametric equations $x = t^2 + 1$, $y = 2t$, $t \geq 0$. The region bounded by C , the x -axis and the lines $x = 2$ and $x = 5$ is rotated about the x -axis. [4]
By first finding the values of t corresponding to $x = 2$ and $x = 5$, find the exact volume of the solid formed.

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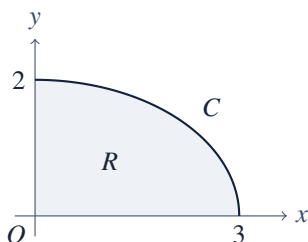
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10. The curve C has parametric equations

$$x = 3 \cos t, \quad y = 2 \sin t, \quad 0 \leq t \leq \frac{\pi}{2}.$$

The region R is bounded by C and the coordinate axes, as shown below.



(a) The region R is rotated about the x -axis. Show that the volume of the solid formed is given by [3]

$$12\pi \int_0^{\frac{\pi}{2}} \sin^3 t \, dt.$$

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(b) Hence find the exact volume of this solid. [2]

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(c) The region R is instead rotated about the y -axis. Find the exact volume of the solid formed. [4]

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(Total for Question 10 is 9 marks)



ROAR-LEVEL CHALLENGE

5 marks

11. The curve C is one arch of a cycloid, with parametric equations

[5]

$$x = t - \sin t, \quad y = 1 - \cos t, \quad 0 \leq t \leq 2\pi.$$

The region bounded by C and the x -axis is rotated through 2π radians about the x -axis.
Show that the volume of the solid formed is $5\pi^2$.

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END OF WORKSHEET

Total: 47 marks